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Australian Football

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Performance Demands of Australian Football

In this section, we introduce Australian football and discuss the critical factors that impact player performance and the relative contribution that physical performance makes to success.

Sport Description and Factors of Winning

Australian football (colloquially known as *Aussie rules*) is a team sport played between two teams of 22 players (18 on-field players and 4 interchange players) with an oval-shaped ball. Despite being indigenous to Australia, the major professional league, the Australian Football League (AFL), ranks among the most attended football leagues in the world. The game is played on a nonstandard-sized oval-shaped natural grass pitch 110 to 155 m wide and 135 to 185 m long with goals at each end (figure 21.1). In the AFL, each quarter is ~30 min (20 min of playing time and ~10 min of time-off), with a 6 min rest period following quarters 1 and 3, and a longer 20 min halftime period between quarters 2 and 3. Eighteen players are allowed on the field at any one time, with the remaining 4 players resting on the interchange bench. The interchange

players may be rotated onto the field to replace an existing player from the same team who leaves the field. A team is permitted to make a number of player interchanges during a match (varied number governed by the AFL).

The ultimate aim is for the team to score more points than the opposition by moving the ball through the goals at their attacking end of the field. Scoring is achieved by kicking the ball between four goal posts positioned 6.4 m apart at each end of the field, with 6 points awarded for a kick through the two middle posts and 1 point awarded for a kick through the outer posts (figure 21.1). Teams score by working collaboratively to maintain possession of the football and moving the ball to their attacking end of the field (forward line) through passing the ball, handballing, or kicking the ball to their teammates. Players are permitted to use their hands to control the ball but are not permitted to run with the ball more than 15 m without bouncing, handballing, or kicking the ball. Between-player body contact is permitted but limited. Opponents are allowed to tackle the player in possession of the ball between the shoulders and knees; push, bump, and block players within 5 m of the football (shepherding); and make contact incidental to a legitimate marking contest (i.e., where players duel to catch a ball kicked to them on the full). In contrast, the aim

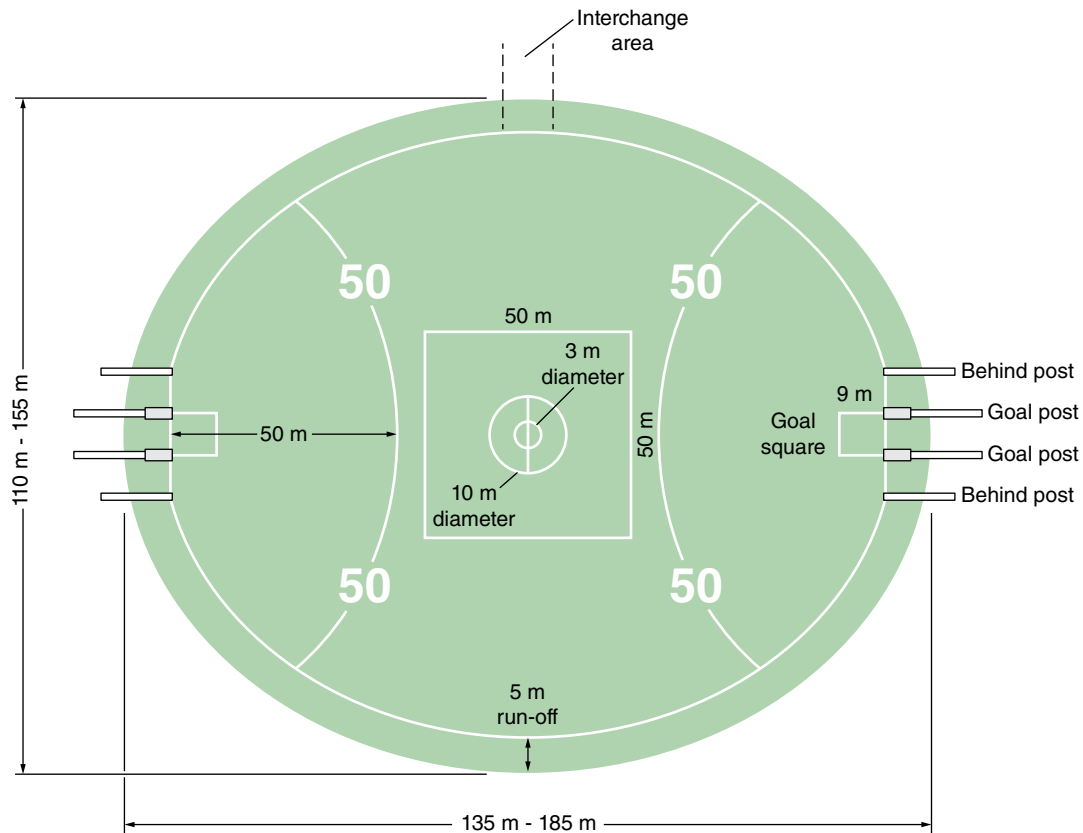


Figure 21.1 Australian football playing field dimensions.

of the team not in possession of the ball is to cause a turnover in ball possession by providing defensive pressure on the ball carrier and players who are likely to receive the ball from that player.

Playing positions have traditionally been classified by the general role that they play such as forwards, midfielders, defenders, and rucks (15, 21), but recently there has been a hybridization of these roles with many players filling many tactical positions often requiring them to rotate through the forward line, midfield, and defensive positions at different periods during a match. The number of strategic positional moves among players on the field is unlimited. This recent change has increased the need for all players to possess high-level intermittent running capacity, speed, and strength to permit them to play in multiple positions on the ground. Despite the increase in hybrid players, recent studies still show distinctive activity profiles and technical skills for various tactical roles (16). While it is important for football players to improve their physical and physiological qualities, technical skills, tactical awareness, and football intelligence are critical contributors to individual and team success (figure 21.2).

Relative Contribution of Physical Performance

Similar to other field-based team sports, individual Australian football player performance is difficult to assess. Indeed, there are innumerable constructs including physical activity capacities, tactical awareness, technical skill involvements, and skill efficiency that contribute to overall performance (figure 21.3). Certainly, high-level performance in professional Australian football relies on well-developed physical, technical, and tactical abilities (39).

It is logical that players develop fitness qualities to cope with the worst-case scenarios (i.e., repeated-sprint bouts, repeated high-intensity bouts, or short periods of intense running) that may appear in match play. It also follows that increased fitness levels are advantageous for facilitating recovery from these highly demanding efforts. Additionally, with the recent reduction and cap on the number of player interchanges permitted during games (which reduces the number of rest periods available during games), it is becoming increasingly more important to prepare highly trained players with the fitness capacities

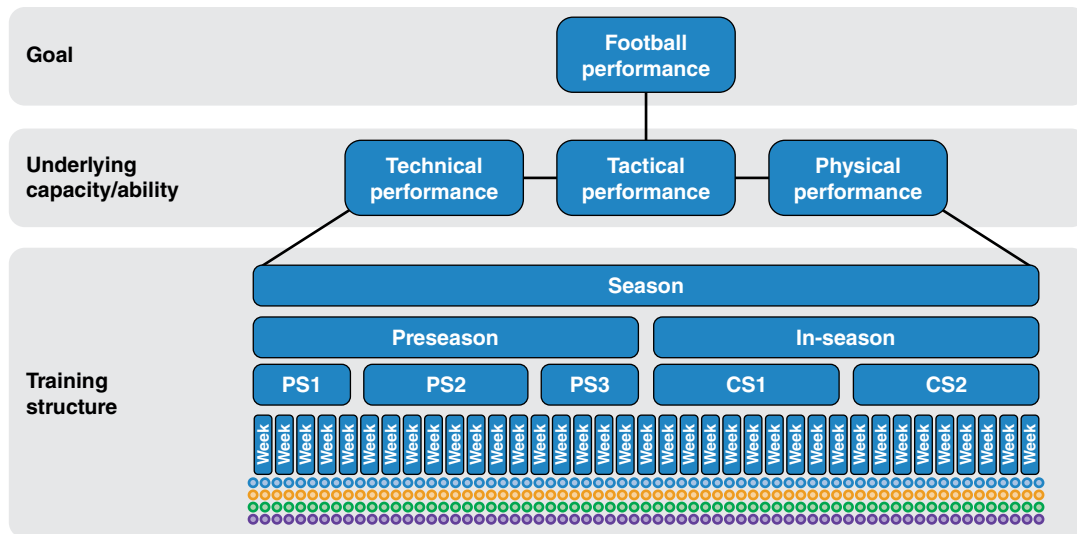


Figure 21.2 The constructs of Australian football performance and the underlying training structure that provides the basis for developing these constructs in professional Australian football. PS: preseason; CS: competition season.



Figure 21.3 The position of the Australian football player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in Australian football, acknowledging that this varies slightly between positions. Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (40%), tactical awareness (25%), and physical capacities (35%) for Australian football success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

to endure the arduous demands of match play for longer. Therefore, high-intensity interval training (HIIT) is an essential aspect of preparing Australian football players.

It is well established that players are required to travel large distances at high average speeds with

significant high-speed running essential to perform well in professional Australian football (15, 16, 26, 41). Similarly, sprinting, accelerations, and repeated high-intensity efforts have also been reported to be important for Australian football performance, with increased requirements at the higher competition

levels (15, 26, 41) compared to subelite (1) or youth (9) competitions. Indeed, highly trained AFL players on average complete $\sim 29 \pm 8$ sprints during match play (15, 41), which is ~ 1 sprint every 3 min. Notably, peak speeds achieved during match play were 29.8 ± 1.7 km/h with peak values of 34.6 km/h. Varley et al. (41) showed that 45% of these sprint efforts were completed with >30 s recovery, showing they occur during intense periods of play and therefore may be important to performance. Professional AFL players have been shown to have a high frequency of high-speed running efforts (i.e., efforts >19.8 km/h, 0.8 ± 0.2 min⁻¹), which is approximately double the rate reported in professional soccer (0.4 ± 0.2 min⁻¹) and rugby league players (0.4 ± 0.2 min⁻¹) (41).

In addition to understanding the average demands of competition, it is also important to have insight into the most difficult periods of activity (or worst-case scenarios) in match play. A recent study reported on the duration-specific running intensities during match play and demonstrated the average speed (i.e., m/min) and peak high-speed relative distances to be greatest when data were analyzed in 1 min periods with midfielders compared to other positions, peaking at 223 ± 35 m/min and 110 ± 35 m/min for 1 min epochs (16). Notably, however, the acceleration and deceleration demands were similar across positions during these periods. Like most sports, these acceleration and deceleration efforts, which are metabolically demanding, occur ~ 4 times more frequently than sprinting. The reason for the increased frequency of acceleration and decelerations is that Australian football players are rarely provided the opportunity to reach maximal velocity due to spatial constraints imposed by field dimensions or opposing players (25). Therefore, the ability to generate speed over a short distance or duration is vital for successful moments in play (i.e., to beat an opponent or to make a break in play).

Both repeated-sprint (RS) and repeated high-intensity effort (RHIE) bouts have been used to classify intense moments in match play and have been suggested to be important for performance. A RS bout was defined as three or more sprint efforts with less than 21 s recovery between efforts (38), while RHIE bouts have been defined as consisting of three or more maximal acceleration, sprint, or collision efforts with less than 21 s recovery (2, 41). Despite general reports that RS and RHIE bouts during match play are important in Australian football performance, few studies have investigated the prevalence of these events in match play. Indeed, Varley

et al. (41) reported that professional Australian football players complete less than one RS bout in each match, suggesting that these are rare events. However, RHIE bouts were more frequent, observed once every ~ 9.5 min, suggesting that this classification might be a more important focus than RS bouts for training and preparation activities. This relatively high number of RHIE bouts in Australian football match play provides insight into the potential physical stress imposed on players during competition.

In summary, there are many constructs of physical performance that have been reported to be important for Australian football performance, including endurance capacity, high-intensity running, sprinting, accelerations and decelerations, as well as collisions, sport-specific skills, and RHIE. Determining the relative contribution of these to Australian football performance is difficult as recent studies highlight the complex interactions between individual players' fitness characteristics, mediating factors around match play (i.e., opponent, draw, time of season, etc.), and other match contextual factors (10, 23, 24, 35, 40, 42) on these variables in match play. Regardless, it is clear that the physical activity demands are high at greater levels of competition and are generally greater in midfielders and hybrid backs and forwards compared to tall forwards, tall backs, and rucks. It is also clear that the activity demands for professional Australian football are high, and players should prepare specifically for these so that they can perform in competitions. However, current understanding also shows that professional Australian football players do not necessarily need to be the fittest athletes, but are required to develop sufficient fitness levels (which are high) to cope with the demands of the match and their opponents and also be able to execute their tactical role and skills efficiently.

Targets of Physical Performance of Australian Football

The physical performance targets of Australian football players as a whole are similar across positions; however, subtle differences exist among midfielders, tallers, and ruckmen. These differences can influence our HIIT targets and therefore must be carefully considered before we choose our training weapons.

Midfield and Hybrid Players

Midfield and hybrid players—mobile forwards and backs—usually have the greatest locomotor and

activity demands in professional Australian football (15, 16). This is likely a result of their nomadic role, which allows them greater pitch space to move in (midfielders) or demand to consistently lead up the ground to make space, or as a marking target before sprinting back to contests of the targeted tall players (for mobile forwards and backs). Accordingly, well-developed anthropometric characteristics, including low levels of body fat and a high proportion of muscle mass, are essential for their performance (3, 4). Similarly, high levels of speed and explosive muscular power are important physical factors for performance at the highest level (4). During match play, depending on their position and tactical strategy, professional players typically cover ~10.2 to 16.8 km (with individual reports of players traveling >17 km in a game). In addition, players execute hundreds of high-intensity actions in the form of high-speed runs (for a total of 1035 ± 385 m >19.9 km/h), accelerations and decelerations, changes of direction, collisions and tackles, and sport-specific skills such as kicking, handballing, marking, contesting with opponents, and engaging in ruck contests. Finally, since the typical playing time is 98 ± 15 min, endurance capacities are critical. Indeed, a well-developed aerobic system is likely to be beneficial, contributing not only to acute high-intensity performance but also to metabolic recovery between the explosive efforts associated with successful match performance. Accordingly, midfield and hybrid players should be exposed to sufficient training volume to develop endurance capacity and repeated bouts of higher-speed running with acceleration demands so they are best prepared for the worst-case scenarios of running requirements on match day.

Talls

Tall forwards are often used as kicking targets in attack and are often marked by similar-sized defensive players from the opposing team (backs). It is often the role of the tall backs to negate the tall forwards and to spoil their marking attempts during air duels, often resulting in jostling and intense body collisions. To counteract this, tall forwards often complete repeated leads to become marking targets during attacking plays, or alternatively move up the field to create space behind them that allows faster and often more agile forwards to run into to create scoring opportunities. Tall backs typically have lower average speed and high speeds during games, likely due to their role to maintain defensive structures

even when the ball progresses forward. Indeed, they may often zone off following a long lead from a tall forward if it is determined that the forward has run into space that is not considered dangerous for an impending attack of the opposing team. Additionally, with the recent cap in player rotations, tall backs tend to have greater match time than other positions, which may explain their lower relative speeds during matches. Due to these demands, training methods for talls should focus on developing the ability to repeat high-intensity efforts including actions such as jostling with opponents and recovering from these bouts.

Ruckmen

The rucks, usually the tallest players on any team, have the specific role of contesting any restart of play. This occurs when the ball is bounced or thrown up by an umpire and the rucks contest with the opposing ruckmen to jump palm or tip off the ball to their surrounding teammates in an attempt to clear the ball into more space. Due to their height and/or marking ability, rucks have as part of their strategic role to position themselves behind the play to become a tall target to cut off rebounding attacks from the opposing team. They also need to arrive at many contests due to their specific role in ball-ups (stop-pages) and boundary throw-ins. Because of this, rucks have few rests on the interchange bench and relatively high playing times. Due to these tactical roles, rucks have lower running speeds and less higher-speed running than other positions. They are also involved in many aerial contests, which results in frequent jumping, colliding, and jostling with opponents. Accordingly, training programs should focus on developing the ability to repeatedly compete in aerial contests, including jostling and making frequent body contact. Additionally, they need to develop the capacity to repeat these efforts over long periods.

Key Weapons, Manipulations, and Surveillance Tools

The weapons we use refer to the HIIT formats applied in our programs. We use specific weapons to target the desired physiological responses and physical adaptations, while the surveillance tools refer to the system and tools that we use to monitor the individual responses to those weapons. In this section, we present the different HIIT weapons that we have used in preparing professional Australian football

players along with their manipulations and our approaches for monitoring training effectiveness.

HIIT Weapons

Due to the vast demands of the competition, it is essential to target all types of HIIT sessions in Australian football (see figure 1.5; Buchheit and Laursen (6)): type 1, metabolic O_2 system; type 2, metabolic O_2 system + neuromuscular; type 3, metabolic O_2 + anaerobic system; type 4, metabolic O_2 + anaerobic and neuromuscular; and type 5, metabolic anaerobic and neuromuscular (7). Similar to most other team sports, the majority of training comes from skill-based training in the form of modified match play, open- and closed-skill training drills, and game-based HIIT (GBHIIT) or small-sided games (SSGs; ~80%, both pre- and in-season). However, short intervals (~10%, both pre- and in-season, for individual top-ups and rehabilitation), RST (~5%, both pre- and in-season, for individual top-ups and players in rehabilitation), long intervals (~5%, preseason only), and rarely SIT (<1%) are also used (figure 21.4). While the specific proportional use of the various HIIT weapons may be different across the various

groups working in Australian football, it would be highly likely that a mixed-methods approach to deploying these weapons is used in all those working at the professional level.

Manipulations of Interval Training Variables

The mode, duration, intensity, time, and recovery characteristics of each HIIT format can be manipulated to elicit desired acute physiological and locomotor responses. The careful manipulation of these variables and placement of HIIT within the physical training programs can be used to prepare professional Australian football players for performance.

HIIT With Long Intervals (Outdoor)

Longer HIIT intervals are regularly applied during the preseason period when there are requirements to develop cardiopulmonary capacity. Accordingly, there is greater emphasis in prescribing these weapons during preseason phases 1 and 2 ($2 \times$ per week with ~72 h between sessions; figure 21.2). These sessions are easily conducted in a controlled manner with players using distances to control speeds. These sessions are usually conducted over interval distances of 800 to 1600 m at speeds of 90% to 100% of maximal aerobic speed (MAS), repeated 3 to 5 times with short breaks (<3 min) (type 4). Active recovery between efforts is applied to players who have greater need to develop cardiovascular fitness or when there are longer intervals between efforts. Since our squads are large (i.e., 40 to 46 players), we divide the larger group into smaller groups according to fitness levels, and these groups pace themselves so that they reach specific cones (usually coded by color) at appropriate times. These drills are usually applied at the end of training sessions once football-specific loadings have been applied, and the drills for individual players are modified for volume according to injury risk and fitness targets. This approach allows coaches to follow their tactical periodization approach and to develop technical and tactical abilities while allowing individual players to meet training load and fitness targets.

Some manipulations of longer-interval training sessions have been achieved by using similar interval work bout durations and work:rest ratios of activity requiring high-intensity intermittent running that represent worst-case scenario activity profiles observed in match play. As an example, we have used match activity data obtained from competition to

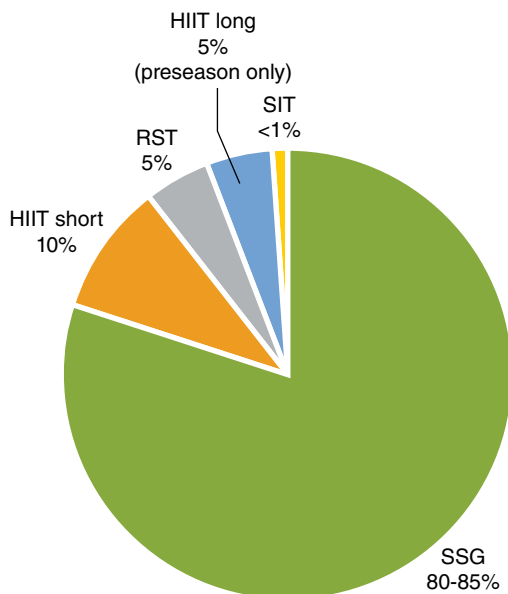


Figure 21.4 Percentage of the different HIIT formats (weapons) used throughout the annual season in Australian football. Game-based HIIT (GBHIIT) or small-sided games (SSGs) make up about 80%, short intervals about 10%, RST about 5%, long intervals about 5% (preseason only), and rarely SIT (<1%).

develop audio files that provide verbal instructions to undertake activity observed during intense periods of match play (i.e., to provide instructions to stand, walk, run, jog, run, fast run, and sprint in sequences observed during intense periods of match play). This audio file is then played over loudspeakers or with a personal iPod and players are tracked with global positioning system (GPS) devices (with heart rate [HR] monitoring). While less controllable than constant velocity HIIT, this approach provides increased movement specificity (e.g., 2 to 4 sprints with accelerations and decelerations in 4 min bouts) and high cardiovascular and anaerobic metabolic stimulus (type 4 targets).

HIIT With Short Intervals

In Australian football, short-interval HIIT is an effective training weapon. The prescriptive variables can be manipulated to carefully control the running speeds, neuromuscular load, and anaerobic contribution during the session to target specific physiologic targets. Indeed, these HIIT formats can be manipulated to meet type 1, 2, 3, or 4 targets. While we use these most regularly during the preseason, they can also be prescribed during the in-season when additional HIIT is required. These sessions are also commonly applied in rehabilitation and form much of the basis of the return-to-sport protocols following injury (depending on the nature of the injury). They are also effective for adding at the end of training sessions to help players meet individual load goals for high-speed running and neuromuscular load.

In the early phases of the preseason, short-interval HIIT sessions that meet target type 1 can be useful for reintroducing interval training after the layoff. For example, 3 sets of 5 intervals of 10 to 15 s work intervals at 100% to 110% MAS with 10 to 15 s passive recovery are well tolerated aerobic sessions. These prescriptions help players avoid reaching high peak speeds and fast accelerations and can be completed after football training without overstressing the neuromuscular system. Different formats can then be used once players adapt to training so that type 2, 3, and 4 targets are met as we move through preseason. For example, we find that once early preseason adaptations have occurred, we commonly apply short-interval HIIT weapons of 15/15 s and 20/20 s at 120% MAS. Players cope well with these interval durations but they tend to have difficulty maintaining running technique during longer-duration short-interval HIIT sessions (espe-

cially at the end of demanding skill-based sessions). Indeed, when developing speed endurance, we rarely have players run at speeds >130% MAS to assist them with maintaining form and developing movement efficiency when experiencing fatigue. We reduce work:rest ratios during short HIIT and use passive recovery to reduce anaerobic load, while increasing both work-interval intensity and the work:rest ratio to increase anaerobic contributions.

Short-interval HIIT can be modified by adding in changes of direction, ball carries and bounces, and position-specific running patterns to increase match specificity and modulate the neuromuscular loading (e.g., move from type 1 to 2 or type 3 to 4). However, careful adjustments need to be taken into account when developing the distances and duration of efforts, as when multiple changes in direction and effort durations are included in a bout, the neuromuscular load can be substantial; thus, significantly reducing the amount of higher speed running completed may be necessary in this situation. We recommend trialing new short HIIT drills on staff or smaller groups of players with stochastic work efforts or changes of direction at various angles before implementing broadly.

We also regularly apply short-interval HIIT weapons to ensure players meet higher-speed running targets during the season. For example, we have set weekly high-speed running exposure targets based on within-individual exposure history, acute:chronic workload ratio, and the period of time between match play. These factors, combined with collective experience from the sport science team and coaching staff, alongside the monitoring data, allow us to titrate training goals and manipulate the training stimulus on an individual level.

Repeated-Sprint Training

Although less common than long- and short-interval HIIT formats, RST weapons can be used in the middle and latter stages of the preseason or during the rehabilitation process in long-term injured players. RST can be useful for developing the ability to do mechanical work with changes of direction, rather than developing higher speeds. Indeed, we often implement RST using a variety of change of direction (COD) with progressions (sometimes with multiple COD in an effort to reflect match running patterns, type 5), differences in starting positions for each effort, and with varying running patterns. We also include football-specific skills in these efforts. For example, we may use stationary, rolling, and contested

starts (i.e., starting while jostling with an opponent) to increase the complexity of the drill. This approach allows players to develop the ability to repeat high-intensity efforts but in a match-specific context. This type of training is completed in low volumes (i.e., 2 or 3 sets of 3 or 4 efforts) with short-effort durations (4 to 8 s). To increase anaerobic responses, sprint effort and recovery durations are increased (type 5), while shorter recovery periods between efforts are used to increase the aerobic stimulus (type 4).

Game-Based High-Intensity Interval Training

GBHIIT forms the large majority of training content in Australian football and is the central focus of any team's preparation strategy. This type of training is made up of closed- and open-skill drills including SSG training (both handball and kicking SSGs) (11, 18, 27). The main objectives of this work are to develop team strategy, optimize player positioning, develop collective cohesion and timing (in offense and defense and transition between these), and manipulate the stimulus by altering a wide range of prescriptive variables. While the prescription of GBHIIT is typically the domain of expert coaches, where the main focus is not to provide an HIIT stimulus, we like to quantify and understand the demands of these activities via GPS, HR, and skill coding to better understand the overall stimulus provided through this training. Closed-skill training drills are easy to manipulate according to traditional HIIT prescription guidelines (6, 7) and can be used to hit all targets (types 1 to 5), but match simulation and SSGs can be more complex. To overcome this, we have quantified the specific training dose and responses to each type of training drill (open and closed) and use this information as part of decision-making when selecting complementary HIIT. We also assess the duration-intensity relationship of various exercise intensity markers in these drills, which allows us to predict the stimulus provided from a training session before it is delivered. This approach allows us to assist coaches in choosing the GBHIIT exercises that meet both technical/tactical and physical goals. However, GBHIIT can be limited, and the typical responses within a group may be variable, and therefore it is not as efficient at targeting specifically desired training outcomes. As such, we complement this training with more tightly constrained HIIT training drills to ensure desired physiological and physical responses are elicited for

each session. After all, a mixed-methods approach to choosing our training methods is likely the best approach for a team sport such as Australian football, as performance in matches requires a wide range of physical abilities and physiological capacities.

While generally of lower intensity than open-skill training drills, closed-skill drills involve kicking and handballing (i.e., line drills) and can also be used to elicit desired HIIT responses through manipulating the drill duration, level of coaching, distance between marker cones, number of lines (i.e., groups), players in each group, etc. For example, cone-to-cone straight-line short-kicking drills (~25 m kick distance with 35 to 45 m run distance) can be prescribed as 5 s efforts (~35 to 45 m of running) with passive rest of about 25 s when there are 5 players in each group and can be used as a form of RST training if completed at ~130% to 160% MAS. These drills can be completed in 3 to 5 sets with short active breaks between each for coaching or drink breaks. Simple alterations can be made by reducing the numbers of players in each group, increasing the work:rest ratio and anaerobic contribution to the drill. We can also modulate the targeted responses during SSGs using different prescriptions. Indeed, similar effects of manipulation of SSG design constraints that have been widely reported in soccer are also apparent in Australian football (22, 33). Specifically, in addition to typical interval bout prescription variables (6, 7), closed-drill design constraints include the player number, pitch size, the number of balls active in a drill, level or extent of opponent pressure, rule modification, as well as coaching feedback to elicit the targeted response for each session (18). In open drills, we can alter rules around the movement path of the ball, role of specific players in drills, as well as positioning or setup at specific parts of the drill. To fully understand the impact of the variation of these prescriptive variables on physical and physiological responses is difficult due to the impact of the number of degrees of freedom of these choices. Nonetheless, there are some rules of thumb that we follow when we prescribe training and fine-tune drill design.

We have found that locomotor and HR responses during match-simulation skill drills best replicate match demands when played on large-pitch areas (i.e., full-sized field drills) (type 4). Moreover, when these are completed with fewer players, the higher speed running, cardiovascular, and kicking demands increase markedly. However, when pitch area is reduced, higher speed running and sprint efforts are

restricted, but accelerations and decelerations increase, resulting in high neuromechanical loading along with increased collision loads.

Match-simulation drills can be used to increase endurance capacity of players when completed in multiple (i.e., 3 to 5) shorter interval bouts (<6 min), but short rest efforts between intervals (<3 min) can alter the neuromuscular and locomotor demands through drill design. To increase the aerobic stimulus, we use longer drill bouts (>8 min) with larger field size, and locomotor demand is increased when there is no inferred pressure placed on the ball carrier. To reduce the running demands but increase acceleration and deceleration demands, we reduce the pitch size, increase player number, or increase the pressure placed on the ball carrier (e.g., handball games with shadow pressure, type 4). To meet type 3 targets, longer bouts of lower space drills at lower speeds can be prescribed. These type 3 drills are better suited to training sessions in which neuromuscular fatigue is being minimized. In contrast, training drills that have greater area available to encourage players to break defensive lines by running and carrying the ball, as well as taking longer leads, promote higher speed running, but can result in greater neuromuscular fatigue and are best suited for in-season to midweek training sessions.

Load Surveillance and Monitoring Tools

With the recent developments of wearable player tracking technologies, it is now a relatively simple task to track the training in both training and match play for all players during every session. Despite the ease of measurement, however, we still pay particular care to methods that we use to collect, treat, and interpret the data we obtain from our players. Additionally, we also aim for a simple approach to our athlete monitoring strategies so that we don't waste precious resources or time in the daily data tsunami (12). We are also careful to follow measures that have established validity, with known levels of reliability (i.e., measurement noise) and with scientific support or strong proof of concept for their use. We also take care to get advice from coaches on what information they value from our monitoring (13).

The goal of monitoring is to predict, measure, and track changes in individual daily locomotor load. We also use the acute/chronic ratio (20) as a heuristic to guide relative risk of any changes in planned training for the immediate and future training sessions. We

also monitor individual players' exposure rates to the number of sprint efforts to check that players regularly exceed >90% of their best match sprint speed during training throughout the season. Again, while this is a simple heuristic without established empirical support, the general intent is to have players achieve these speeds in two separate training sessions each week to protect against neuromuscular detraining. HR is also monitored during most training sessions that include a metabolic component (i.e., skills and football-related and HIIT-conditioning sessions), and we collect RPEs after all sessions. HR is used to track the cardiorespiratory response, and we pay specific attention to the mean heart rate but also the time spent in higher heart rate zones (i.e., >90 individual HRmax) over time. This is particularly useful in the preseason period or following a long-term injury layoff when we are aiming to develop cardiovascular fitness within individual players. The session-RPE training load is used to assess the overall load experienced by the players, and it can be analyzed by training type and assessed for temporal changes within individuals and teams (see also chapter 8). The session-RPE method has been validated for Australian football (36) and is likely the most valid measure of internal training load. The session-RPE load and intensity measures are interpreted in the context of planned training load and other measures of external training load determined from GPS (i.e., locomotor loads such as total and absolute distances and high-speed running) and other microsenors (i.e., accelerometers).

Training Status Surveillance and Monitoring Tools

Our athlete monitoring system has been developed over more than 20 years and its practical application and theoretical basis has been described elsewhere (14). We have tried most monitoring variables (including expensive and invasive blood measures, heart rate variability, a variety of neuromuscular screening measures, and countless markers derived from GPS technology) and have found that most variables were unsuitable for *our* environment, where we monitor ~46 young males daily in a time-poor environment. Following considerable trial and error (lots of time and *many errors!*), we have arrived at a simple system that includes mostly non- or low-invasive measures such as session-RPE (19), GPS-load monitoring, wellness (8, 29), exercise HR

(fitness) (5, 8), and simple neuromuscular screening tests such as groin squeeze and clinical screening feedback from medical staff. The most critical aspect to the success of this system is the buy-in from players and coaching staff, and therefore we require involvement from key decision-making stakeholders, including senior coaching staff (senior coach or senior assistant coach), performance staff, and medical staff. To monitor an individual player’s readiness to perform and/or to assess changes in fitness or training adaptation, we monitor the locomotor

(GPS-related), HR, and RPE responses to either standardized SSGs (32), match simulations, or submaximal shuttle runs (8). In our analysis, we examine the relationship between changes in the internal:external load ratio, where the greater the load/min and lower the HR and RPE indicate an improved status. We then often use these individual responses (in conjunction with other wellness and neuromuscular screening measures) to determine future training or treatment prescription. An example of the team session-RPE loading is shown in figure 21.5.

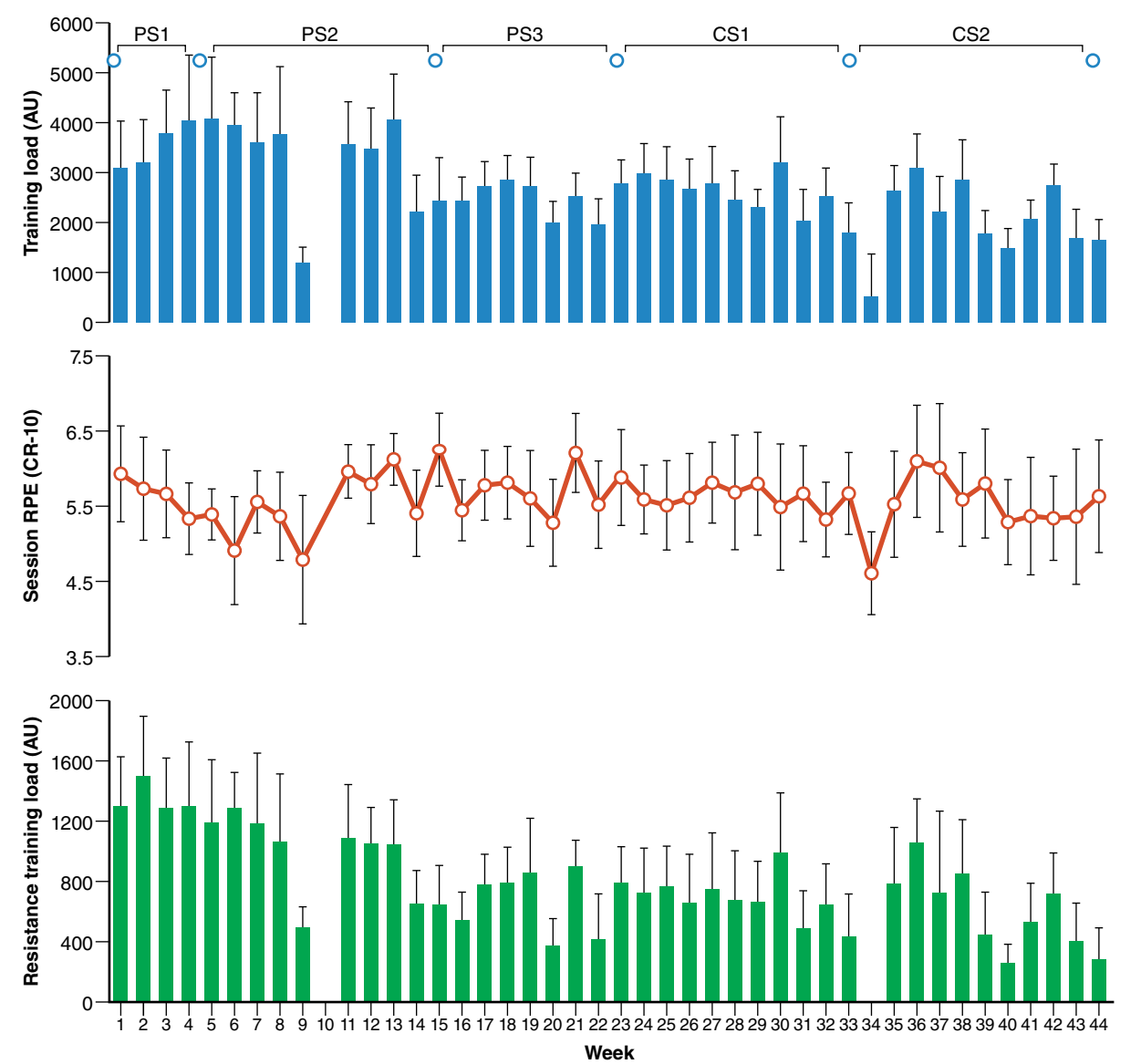


Figure 21.5 Sample team session-RPE loads for overall training (top panel), weekly training intensity (middle panel), and resistance training (bottom panel) during a professional Australian Football League season. PS: preseason; CS: competition season.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in Australian football may be largely determined by various controllable and uncontrollable factors, before proper progression and periodization can be appropriately implemented. This section discusses some of the nuances of training program design and implementation of HIIT in Australian football.

Controllable and Uncontrollable Factors

In fulltime professional Australian football, there are numerous training and nontraining factors that place time constraints on training scheduling. Outside of match time and location (both uncontrollable), the strongest influence on program design is the coaching philosophy, which places a high priority on technical and tactical development. Therefore, the most viable way to introduce HIIT into a program is to add these different HIIT components to existing technical and tactical sessions. To achieve this, careful monitoring is required to understand the precise nature of these sessions. Once this is achieved, we can assist coaches with developing the timing and dosing of specific training, and then we can plan any additional HIIT requirements around these. Through this approach, coaches are able to deliver the technical and tactical training required to meet performance objectives, while also being able to provide appropriate physical training.

Program Structure and Progression

In contrast to popular theory, traditional periodization models of cyclical loading and unloading are uncommon in professional Australian football. The preseason period in professional Australian football is relatively long compared to most other team sports. The preseason period can be 15 to 19 wk (including preseason competition) and is usually divided into 2 or 3 shorter macrocycles with short rest periods between each (figure 21.2). In the first macrocycle, the focus is transitioning the players back to football training after ~2 mo of self-directed off-season training. Accordingly, caution is taken to provide progressive stimulus overload (in duration and intensity), but not to the levels that increase injury risk. We have found this to be a period of higher injury risk, especially when new technical and tactical training

drills are being introduced to the program and with an increased variation in the training status of the players returning from the off-season period (see the case study at the end of the chapter). Additionally, since players do not tend to complete a high load of intense off-line or agility training during this introductory period, we control kicking loads by manipulation of pitch size (i.e., smaller pitch area for fewer shorter kicks) and direct instruction and carefully reintroduce high-intensity actions. We also focus on type 1 HIIT targets during the early phase to avoid overloading the neuromuscular system (e.g., short intervals at 80% to 90% MAS).

In the second preseason macrocycle, the heaviest training with the greatest running volumes and resistance loads is completed (figure 21.6). Due to the lack of match play, this second phase provides the best opportunity to program increased volumes of HIIT. During this period, there is a large focus on technical and tactical training but with significant portions at high intensity. A sample weekly training schedule for the preseason period is shown in table 21.1. We aim our program to touch on all targeted qualities during the week, but with emphasis changes during the preseason. General principles are that the interval efforts should generally be reduced from longer to shorter, interval speeds increase, and the recovery time between efforts is reduced as this phase progresses. We aim to provide two sessions per week (e.g., Monday and Friday) that provide greater higher speed running distances and peak speeds (i.e., type 4 targets) and another midweek HIIT stimulus with high aerobic load but with moderate levels of higher speed running distances and lower peak speeds (type 3 targets).

An important focus for the preseason is developing a robust training base, and this is achieved by developing training continuity. Indeed, recent studies have suggested that players who complete a greater proportion of preseason training also endure higher training loads, which results in greater availability for training and selection during the competitive phase of the season (31). Finally, the third phase of preseason training is in the introduction of preseason practice matches. Clearly during this period there is a shift in programming for training to reflect the in-season training schedule, with lower training doses and decreased variability in dose; however, the emphasis for some players may be on developing physical characteristics if they have not yet been developed.

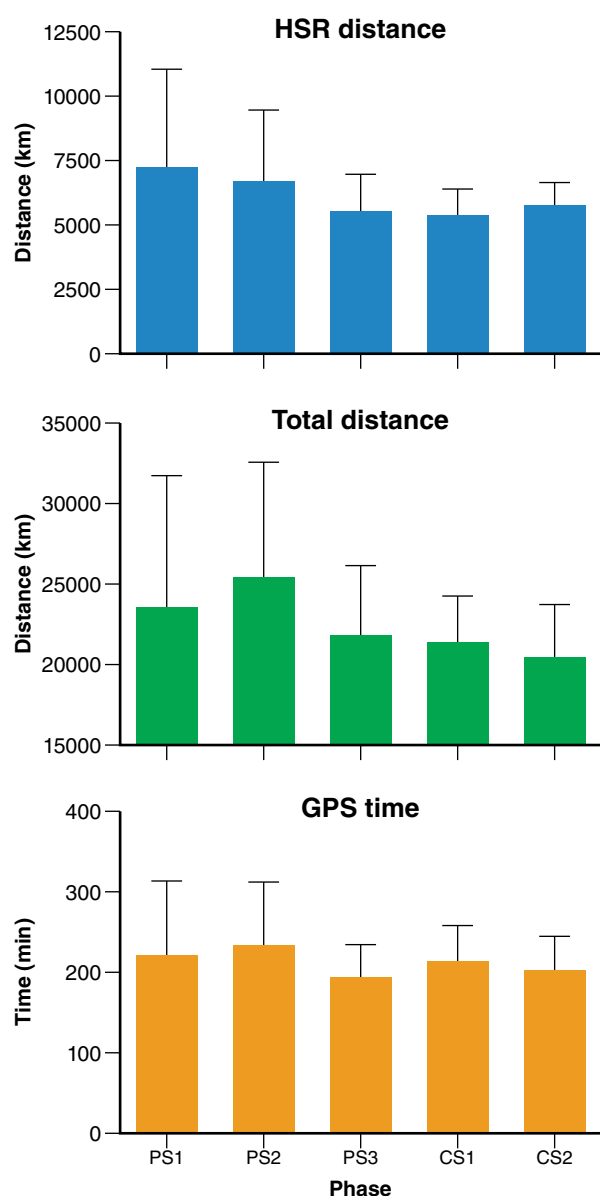


Figure 21.6 Sample GPS-derived weekly values for higher speed running (top panel), total distance (middle panel), and training duration (bottom panel) in a professional Australian Football League season. PS: preseason; CS: competition season.

During the in-season period, the periodization structure is focused on recovering from the previous match and then tapering for the next match. One priority for periodizing training during the in-season is to keep consistency in the weekly training schedule and training load. This can be challenging, especially due to the varying number of days between

matches (i.e., 5 to 12 d). Accordingly, we keep a consistent structure in the weekly in-season program for the common between-match periods and this is rarely modified (table 21.2). The general goal of this structure is to allow for recovery from previous matches, to provide a stimulus between games, and then to allow for a short taper or recovery period prior to the next game. The focus of HIIT during the in-season is to complement the technical and tactical training, and we select HIIT formats that touch on the primary physiological target each week. The weekly match provides the greatest HIIT stimulus and the remainder of training is planned around this stimulus (i.e., to recover and taper before each match). Accordingly, player load management is critical and control of training dose is essential to allow for HIIT. HIIT weapons applied early in the in-season training week are targeted toward type 1 (low volume, recovery, and off-legs) and moderate volume type 3 while type 3 and 4 formats are applied ≥ 72 h after match play (i.e., midweek) after neuromuscular recovery has occurred from the previous match.

A recent case study from the AFL shows decreases in training load and intensity and increases in the proportion of recovery activities during the in-season period (30). This approach to training periodization is supported by laboratory research showing that physical performance during prolonged, high-intensity intermittent exercise is improved when lower training doses are completed between matches completed 5 d apart (37). Additionally, during the in-season, ~50% to 55% of weekly running load is obtained during games, whereas the remainder is obtained by technical and tactical training (34). These data show that there are few opportunities to add isolated HIIT sessions into a weekly schedule. This practical limitation can result in players not receiving adequate stimulus from HIIT, which may increase injury risk (17, 28). Therefore, to mitigate this risk, we ensure that adequate high-speed running and sprint efforts are completed, and we top up training with training drills (e.g., end of warm-ups or SSGs or technical and tactical drills). This approach can protect players from injury and prevent detraining these capacities.

Incorporation of HIIT

In our experience, the primary driver of training periodization and load prescription for professional

Table 21.1 Sample Preseason Training Schedule

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
a.m.	Warm-up Submax HR test Speed technique Skills Recovery (Hydro)	Lower weights Physio treatment/ massage	Warm-up Speed technique Skills Recovery (Hydro)	Off	Warm-up Speed technique Skills Recovery (Hydro)	Individually targeted sessions: • Cross-training • Weights • HIIT running	Off
	Lunch	Lunch	Lunch		Lunch		
p.m.	Weights Physio treatment/ massage	Off	Weights Physio treatment/ massage	Off	Weights Physio treatment/ massage	Off	Off

Table 21.2 Sample In-Season Training Schedule

	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
a.m.	Off	Sleep in Individual recovery	Screening	Warm-up/ activation/speed technique Skills	Off	Warm-up/ activation/speed technique Skills	Warm-up/ activation/ speed technique Captain run
p.m.	Game Post-game recovery	Off	Recovery rotations (massage, hydrotherapy, mobility)	Weights Physio treatment/ massage	Off	Weights Physio treatment/ massage	Off

Australian football comes from the head coach in collaboration with high performance and medical staff. Similar to most team sports, a technical and tactical emphasis in Australian football is of most importance (figure 21.3) and is the first aspect of a training program prescribed. Once the coaches have planned the football-focused technical and tactical training sessions, we then develop team and individual player training plans that may incorporate HIIT. The session planning is done to meet the individual goals for training load (i.e., distances, high-speed running, sprinting distances, and acceleration efforts). To achieve this, we use a bespoke training database system to estimate training dose provided by the training drills planned by the coach. This is achieved by estimating loads (i.e., total distances, higher speed running, sprint number, and accelerations based on historical data) described here: <https://vimeo.com/161594888>. We then program additional training (i.e., HIIT) around this. If required, we usually incorporate HIIT into training sessions both at the team and individual levels at the end of the session.

When planning HIIT sessions, we consider several factors including the recent training completed, the well-being status of the athlete, the athlete's training goals, and the planned training stimulus during the weekly microcycle. Various targets are selected depending on the needs of the individual or team. We also follow several rules of thumb such as planning to avoid on-legs HIIT sessions in the first 48 hours following matches, as this may compromise recovery and subsequent performance. We also avoid programming high-speed HIIT sessions following fatiguing lower-body resistance training sessions or when players report increased levels of lower-body soreness. We also consider the demands of the other training in which HIIT is incorporated and the loading patterns within the weekly cycle, as these factors influence the metabolic and neuromuscular responses to HIIT (6, 7). In reality, the positioning of HIIT sessions within a training day or session is largely determined in consultation with coaches. With the exception of planned periods during the preseason, coaches are often reticent to train skills following heavily fatiguing HIIT, and as a consequence, isolated

HIIT sessions are mostly conducted after main skill-focused training. However, we are also careful to avoid HIIT sessions in which peak speeds are reached at the end of sessions as injury risk may be increased. However, we are also mindful of the protective effect of high-speed sessions so these are programmed around periods when players have less acute residual fatigue and muscle soreness. Indeed, we often plan higher speed days twice a week and demanding sessions with lower speed stimulus between these sessions. Carefully selecting skill-based training drills and manipulating the constraints of training drills described in the section above achieve this. Finally, when heavy HIIT sessions are programmed, these are often positioned in the weekly microcycle immediately prior to days of rest to allow players the best chance to recover from residual fatigue and soreness that may result from these sessions.

In-season programming is largely dictated by the competition draw and travel requirements. Other factors such as recent and acute training loads and expected match difficulty can also influence decision-making. Similar to preseason loading structures, we attempt to maintain a level of consistency in the nature and frequency of HIIT stimulus. During the in-season, it is vitally important to ensure players are sufficiently loaded to maintain their fitness and be prepared for performance (i.e., matches). This requires a highly individualized HIIT prescription, using a combination of different HIIT types and formats, but we avoid on-legs HIIT in the 48 to 72 h post-match.

Sample Training Programs

As discussed, we program HIIT sessions 3 or 4 times per week in preseason depending on individual needs (i.e., athlete's training age, current physical fitness, and expected competition demands). For players with nonmodified training programs, we attempt to provide some level of polarization in the HIIT stimulus within a typical week. Early in preseason, a Monday to Friday HIIT session may consist of a mixed-HIIT format running with a variety of acceleration and deceleration and speed work (as a part of warm-up); a selection of short (15 to 25 s) efforts at 100% to 120% MAS, for 5 to 10 min of work, types 3 and 4; and longer intervals (2 to 6 min efforts) for 4 to 8 bouts at 80% to 100% MAS, type 3, with ~4 to 5 km per session from high-intensity training. The interval effort durations are usually over longer distances to allow an increase in higher

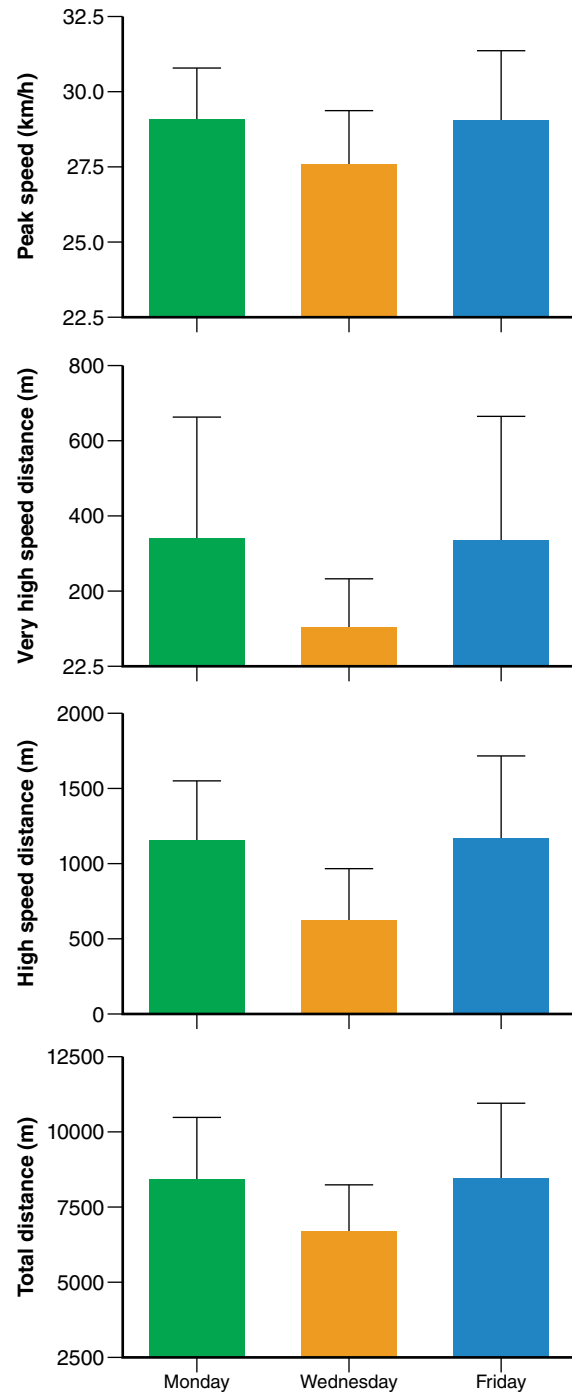


Figure 21.7 Locomotor data for main preseason sessions involving HIIT for peak speed (km/h; top panel), very high-speed running distances (>25 km/h; upper middle panel), high-speed running distance (>19.9 km/h; lower middle panel), and total distance (bottom panel).

speed running and the achievement of high peak speeds. The Wednesday session is a shorter HIIT session and may involve short-interval formats with reduced rest periods between bouts and role-specific run conditioning drills. This results in reduced higher speed running and lower peak speeds for these sessions compared with the Monday to Friday sessions (type 4 to type 3) (figure 21.7). As we move toward the season, there is a tendency to use shorter interval durations, with reduced rest periods between efforts. We also tend to develop position-specific drills that reflect position-specific running patterns. For example, hybrids, forwards, and backs develop RHIE and ability to hold high speeds, while midfielders

work on repeated high-intensity efforts, but with greater focus on acceleration and deceleration with less space to move in.

While we appreciate there are a number of different ways we can skin the cat to prepare players for the season, this approach allows for HIIT to be manipulated based on the stimulus provided from technical and tactical training (i.e., the coach's main priority). During the in-season, the opportunity for HIIT is limited and is mostly achieved through specific GBHIIT. However, more generic HIIT conditioning can be used to top up high-speed running, accelerated running, and exposure to near-peak sprinting speed.

CASE STUDY

This case study is more a story about when things go wrong and the consequences of periodization errors that can arise with HIIT. The events we describe took place relatively early in our careers and, despite the negative consequences, the experience provided an invaluable learning experience for all involved.

We had just completed the first phase of preseason training and the players had returned from their two-week Christmas break in great shape and good spirits. It was the third main training day after our break, and the players looked fresh and had been performing well. They had coped well with two heavy training sessions in the past four days. The first part of the morning plan was a 2 km trial, and many players ran personal best times. We were quite bullish about the state of the players and following the great fitness test results the expectation for a good quality session was high. On the back of these results, we planned a long football-focused training session that included high-intensity technical and tactical training and match-simulation exercises. It was a hot day (93 °F/33.9 °C).

Probably due to the combined effects of residual fatigue from the time trial and the hot environmental conditions, the skill execution of the players was poor during their skills session. Our training drills weren't being completed to the expected standard (intensity and skill errors), and the coaches were becoming frustrated. As a consequence of the poor-quality session, the players were instructed to repeat the drills for longer periods of time until they met expected standards. Ultimately, the planned skills session ended up running much longer than we had planned (47% longer!). And not only that, the day wasn't finished yet.

With coaches still frustrated from the poor skills session, the players were directed to perform a character-building grappling session. This session consisted of about 30 min of technically and physically demanding grappling and body contact work. It was again a very intense session and emotions were high, to the level that one player actually fractured a clavicle! Finally, after a 1 h break for recovery and fueling, the players finished the day with their usual 45 min resistance training session.

It was obvious to the staff that the session had cooked our players. Despite this, the session was explained as being a character-building one, and we quickly moved on to plan the next session. While we were aware that we'd broken a simple rule of training, we were confident we could recover from the session and began planning the next required piece of the programming puzzle.

As might now be expected, the next day's monitoring data clearly revealed that the players hadn't recovered well. The players reported the highest soreness and fatigue levels for the preseason. Our initial plan was to have a longer type 4 HIIT interval session planned for the day, and as a result, we decided to reduce the volume and effort distance so as to reduce the loads and associated injury risk. But even this wasn't enough. Even with these modifications, two players finished the session with calf injuries, while another sprained his toe ligaments! It was a disaster.

Fundamentally, we broke a basic rule of training. We had given our players a large spike in training load, and despite being aware of this, we backed it up with an HIIT session the next day. Retrospectively, this training error is obvious, but at the time we thought we had handled it appropriately.

That training error had a significant effect on the rest of our preseason. We became extra cautious with managing player availability in the weeks following, and as a result we removed HIIT run sessions from our program for the next 3 wk. This allowed us to modify training without foregoing football-related training. We did this to ensure that we had an injury-free playing roster before the season started. But the knock-on consequence of this overly cautious approach was that we weren't able to expose our players adequately so as to prepare them for the arduous AFL season.

Again in retrospect, it is not surprising that we didn't perform as well as we had expected that year. While it is difficult to directly blame our season results on one or two training errors made 2 mo before the season actually started, we are confident that it contributed, and highlights the fall-on effects from inappropriate training in elite sport. We walk a fine line sometimes. Even to this day we regret the poor load management that occurred, as we should have known better. But we did learn a valuable lesson in that great care should be taken at all stages of the preparation process. We are now well aware that even a few small training errors and lack of thought in training delivery can have substantial long-term consequences. We learned that in elite sport, we can't take our eyes off the ball!

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